Calculating Student Performance From Various Student Factors

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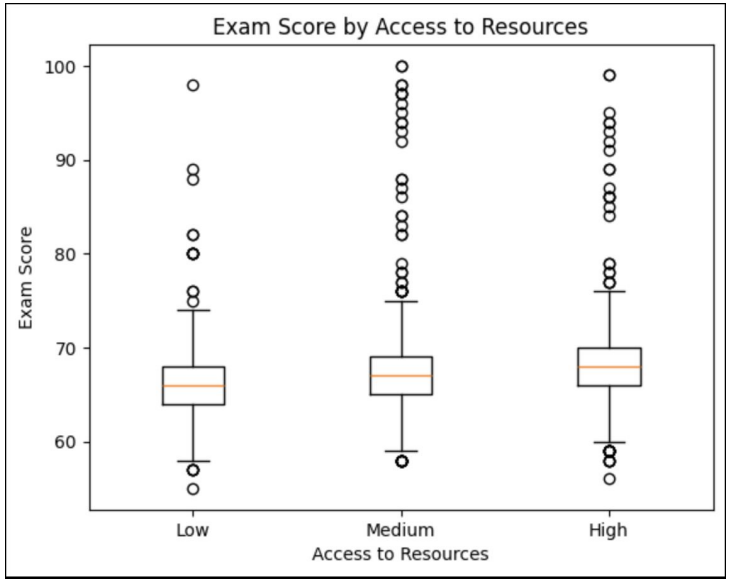
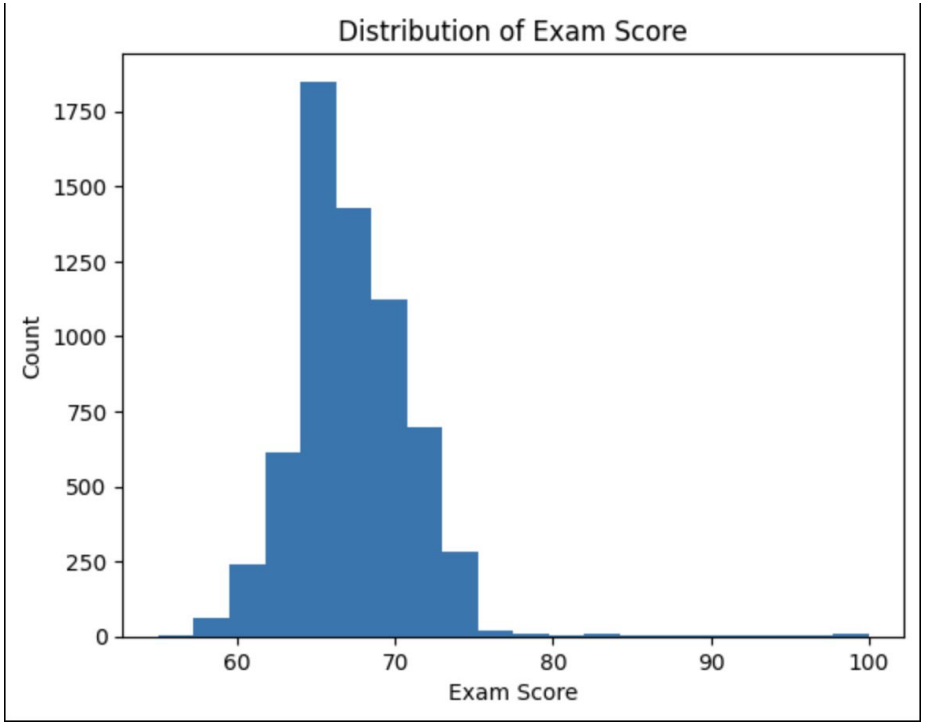
Problem Description

1. Attempt to predict exam scores from student measures.
2. Clean, prep, and then compare 2 models
3. Find best performing model and asses real world error

Data Preparation

1. Initial cleaning
2. Handling missing values
3. Outlier handling





Feature Engineering and Pipeline

1. Data split into train, validation, and test
2. Numerical features scaled
3. Categorical values encoded
4. Highly skewed variables log transformed

Modeling

1. Comparing SVM vs XGBoost

SVM

Key properties	Hyperparameters tuned	Best parameters
• Effective in high-dimensional spaces • Robust to overfitting with appropriate regularization • Sensitive to feature scaling	• C (regularization strength) • gamma • kernel	• C = 10 • kernel = linear • gamma = scale

XGBoost

Key properties	Hyperparameters tuned	Best parameters
• Captures complex nonlinear relationships • Handles interactions between variables • Often achieves strong performance on tabular data	• learning rate • tree depth • number of trees	• learning rate 0.05 • max depth 3 • 400 trees

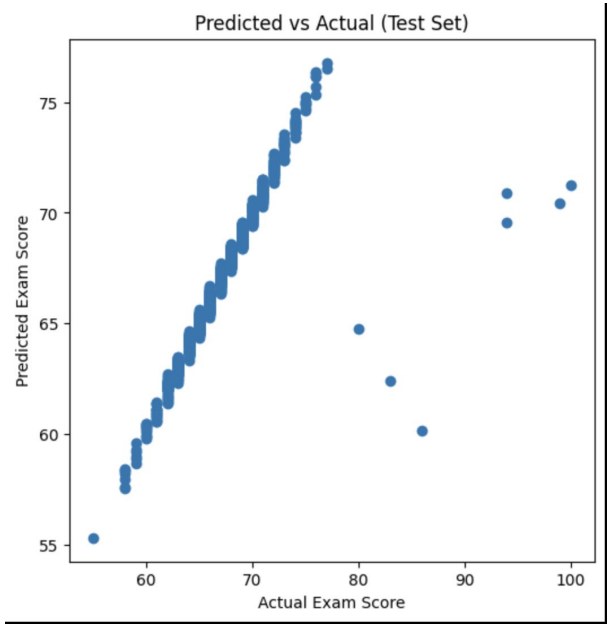
Model Evaluation

Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R2 Score
• Average absolute difference between predicted and actual scores • Easy to interpret in exam score units	• Penalizes large errors more heavily • Useful for detecting large prediction mistakes	• Measures proportion of variance explained by the model • Higher values indicate better predictive performance

SVM	XGBoost
• MAE: ~0.48 • RMSE: ~2.27 • R2: ~0.67	• MAE: ~0.69 • RMSE: ~2.31 • R2: ~0.66

Final Test Results

SVM (train + val)	MAE ~0.44	RMSE ~2.09	R2 ~0.74



Conclusion

1. Exam scores can be predicted with reasonable accuracy using machine learning models
2. Educational datasets can reveal meaningful patterns with regards to academic performance
3. These models can be used to further improve the learning experience of students by identifying under-performers early and taking the right actions to correct the course